

Document number	P0792R1
Date	2017-11-26
Reply-to	Vittorio Romeo < vittorio.romeo@outlook.com (mailto:vittorio.romeo@outlook.com)>
Audience	Library Evolution Working Group
Project	ISO JTC1/SC22/WG21: Programming Language C++

function_ref: a non-owning reference to a Callable

Abstract

This paper proposes the addition of `function_ref<R(Args...)>` to the Standard Library, a “*vocabulary type*” for non-owning references to `Callable` objects.

Changelog and polls

From P0792R0 to P0792R1

Changes

- Removed empty state and comparisons with `nullptr`;
- Removed default constructor;
- Added support for `noexcept` and `const`-qualified function signatures (*these are propagated to `function_ref::operator()`*);
- Added deduction guides for function pointers and arbitrary callable objects with well-formed `&remove_reference_t<F>::operator()`;
- Added two new bullet points to “Open questions”;

- Added “Example implementation”;
- Added “Feature test macro”;
- Removed noexcept from constructor and assignment.

Semantics: pointer versus reference

option 1

function_ref, non-nullable, not default constructible

option 2

function_ptr, nullable, default constructible

We want 1 and 2

SF F N A SA

1 2 8 3 6

ref vs ptr

SR R N P SP

6 5 2 5 0

The poll above clearly shows that the desired direction for `function_ref` is towards a *non nullable, non default-constructible* reference type. This revision (P0792R2) removes the “empty state” and default constructibility from the proposed `function_ref`. If those semantics are required by users, they can trivially wrap `function_ref` into an `std::optional<function_ref<...>>`.

target and target_type

We want target and target-type (consistent with `std::function`) if they have no overhead

Unanimous consent

We want target and target-type (consistent with `std::function`) even though they have overhead

SF F N A SA

0 0 1 9 4

I am not sure whether `target` and `target_type` can be implemented without introducing overhead. I seek the guidance of the committee or any interested reader to figure that out. If they require overhead, I agree with the poll: they will be left out of the proposal.

Table of contents

- [Overview](#)
- [Motivating example](#)
- [Impact on the Standard](#)
- [Alternatives](#)
- [Synopsis](#)
- [Specification](#)
- [Feature test macro](#)
- [Example implementation](#)
- [Existing practice](#)
- [Possible issues](#)
- [Open questions](#)
- [Bikeshedding](#)
- [Acknowledgments](#)
- [References](#)

Overview

Since the advent of C++11 writing more functional code has become easier: functional programming patterns and idioms have become powerful additions to the C++ developer’s toolbox. “**Higher-order functions**” are one of the key ideas of the functional paradigm - in short, they are functions that take functions as arguments and/or return functions as results.

The need of referring to an existing `Callable` object comes up often when writing functional C++ code, but the Standard Library unfortunately doesn’t provide a flexible facility that allows to do so. Let’s consider the existing utilities:

- **Pointers to functions** are only useful when the entity they refer to is stateless (*i.e. a non-member function or a capture-less lambda*), but they are cumbersome to use otherwise. Fully supporting the `Callable` concept requires also explicitly dealing with **pointers to member functions** and **pointers to data members**.
- `std::function` seamlessly works with `Callable` objects, but it’s a “*general-purpose polymorphic function wrapper*” that may introduce unnecessary overhead and that **owns** the `Callable` it stores. `std::function` is a great choice when an owning type-erased wrapper is required, but it’s often abused when its ownership semantics and its flexibility are not required.

- Note that when `std::function` is constructed/assigned with a `std::reference_wrapper` to a `Callable`, it has reference semantics.
- Another limitation of `std::function` is the fact that the stored `Callable` must be `CopyConstructible`.
- **Templates** can be used to avoid unnecessary costs and to uniformly handle any `Callable` object, but they are hard to constrain to a particular signature and force code to be defined in headers.

This paper proposes the introduction of a new `function_ref` class template, which is akin to `std::string_view`. This paper describes `function_ref` as a **non-owning lightweight wrapper** over any `Callable` object.

Motivating example

Here's one example use case that benefits from *higher-order functions*: a `retry(n, f)` function that attempts to synchronously call `f` up to `n` times until success. This example might model the real-world scenario of repeatedly querying a flaky web service.

```
struct payload { /* ... */ };

// Repeatedly invokes `action` up to `times` repetitions.
// Immediately returns if `action` returns a valid `payload`.
// Returns `std::nullopt` otherwise.
std::optional<payload> retry(std::size_t times, /* ????? */ action);
```

The passed-in action should be a `Callable` which takes no arguments and returns `std::optional<payload>`. Let's see how `retry` can be implemented with various techniques:

- Using *pointers to functions*:

```
std::optional<payload> retry(std::size_t times,
                           std::optional<payload>(*action)())
{
    /* ... */
}
```

([on godbolt.org](https://godbolt.org/g/UQbZYp)) (<https://godbolt.org/g/UQbZYp>)

- **Advantages:**
 - Easy to implement: no need to use a template or any explicit constraint (e.g. `std::enable_if_t<...>`). The type of the pointer specifies exactly which functions can be passed, no extra constraints are required.
 - Minimal overhead: no allocations, no exceptions, and `action` is as big as a pointer.

- Modern compilers are able to completely inline the call to action, producing optimal assembly.

- **Drawbacks:**

- This technique doesn't support stateful Callable objects.

- Using a template:

```
template <typename F>
auto retry(std::size_t times, F&& action)
    -> std::enable_if_t<std::is_invocable_r_v<std::optional<payload>, F&&>,
                        std::optional<payload>>
{
    /* ... */
}
```

(on [godbolt.org](https://godbolt.org/g/AGikkz)) (<https://godbolt.org/g/AGikkz>)

- **Advantages:**

- Supports arbitrary Callable objects, such as stateful closures.
- Zero-overhead: no allocations, no exceptions, no indirections.

- **Drawbacks:**

- Harder to implement and less readable: users must use `std::enable_if_t` and `std::is_invocable_r_v` to ensure that `action`'s signature is properly constrained.
- `retry` must be defined in a header file. This might be undesirable when trying to minimize compilation times.

- Using `std::function`:

```
std::optional<payload> retry(std::size_t times,
                            std::function<std::optional<payload>()> action)
{
    /* ... */
}
```

(on [godbolt.org](https://godbolt.org/g/t9FH9b)) (<https://godbolt.org/g/t9FH9b>)

- **Advantages:**

- Supports arbitrary Callable objects, such as stateful closures.
- Easy to implement: no need to use a template or any explicit constraint. The type fully constrains what can be passed.

- **Drawbacks:**

- Unclear ownership semantics: `action` might either own the stored `Callable`, or just refer to an existing `Callable` if initialized with a `std::reference_wrapper`.
- Can potentially have significant overhead:
 - Even though the implementation makes use of SBO (*small buffer optimization*), `std::function` might allocate if the stored object is large enough. This requires one extra branch on construction/assignment, one potential dynamic allocation, and makes `action` as big as the size of the internal buffer.
 - If the implementation doesn't make use of SBO, `std::function` will always allocate on construction/assignment.
 - Modern compilers are not able to inline `std::function`, often resulting in very poor assembly compared to the previously mentioned techniques.
- Mandatory use of exceptions: `std::function` might throw if an allocation fails, and throws `std::bad_function_call` if it's invoked while unset.

- Using the proposed `function_ref`:

```
std::optional<payload> retry(std::size_t times,
                           function_ref<std::optional<payload>()> action)
{
    /* ... */
}
```

([on godbolt.org](https://godbolt.org/gDvWKVH)) (<https://godbolt.org/gDvWKVH>)

- **Advantages:**

- Supports arbitrary `Callable` objects, such as stateful closures.
- Easy to implement: no need to use a `template` or any constraint. The type fully constrains what can be passed.
- Clear ownership semantics: `action` is a **non-owning** reference to an existing `Callable`.
- Small overhead: no allocations, no exceptions, and `action` is as big as two pointers.
 - Modern compilers are able to completely inline the call to `action`, producing optimal assembly.

Impact on the Standard

This proposal is a pure library extension. It does not require changes to any existing part of the Standard.

Alternatives

The only existing viable alternative to `function_ref` currently is `std::function + std::reference_wrapper`. The Standard guarantees that when a `std::reference_wrapper` is used to construct/assign to a `std::function` no allocations will occur and no exceptions will be thrown.

Using `std::function` for non-owning references is suboptimal for various reasons.

1. The ownership semantics of a `std::function` are unclear - they change depending on whether or not the `std::function` was constructed/assigned with a `std::reference_wrapper`.

```
void foo(std::function<void()> f);  
// `f` could be referring to an existing Callable, or could own one.  
  
void bar(function_ref<void()> f);  
// `f` unambiguously is a non-owning reference to an existing Callable.
```

2. This technique doesn't work with temporaries. This is a huge drawback as it prevents stateful temporary lambdas from being passed as callbacks.

```
void foo(std::function<void()> f);  
  
int main()  
{  
    int x = 0;  
    foo(std::ref([&x]{ ++x; })); // does not compile  
}
```

(on godbolt.org) (<https://godbolt.org/g/DPQ7ku>)

The code above doesn't compile, as `std::ref` only accepts non-const lvalue references (*additionally, `std::cref` is explicitly deleted for rvalue references*). Avoiding the use of `std::ref` breaks the guarantee that `f` won't allocate or throw an exception on construction.

3. `std::function` is harder for compilers to optimize compared to the proposed `function_ref`. This is true due to various reasons:

- `std::function` can allocate and/or throw exceptions on construction and/or assignment.
- `std::function` might use SBO, which could require an additional branch during construction/assignment, make inlining more difficult, and unnecessarily increase memory usage.

Rough benchmarks comparing the generated assembly of a *`std::function` parameter* and a *`function_ref` parameter* against a *template parameter* show that:

- `std::function`, on average, generates approximately 5x more assembly than a template parameter.
- `function_ref`, on average, generates approximately 1.5x more assembly than a template parameter.

A description of the benchmarking techniques used and the full results can be found on my article “*passing functions to functions*”¹.

Synopsis

```
namespace std
{
    template <typename Signature>
    class function_ref
    {
    public:
        constexpr function_ref(const function_ref&) noexcept;

        template <typename F>
        constexpr function_ref(F&&);

        constexpr function_ref& operator=(const function_ref&) noexcept;

        template <typename F>
        constexpr function_ref& operator=(F&&);

        constexpr void swap(function_ref&) noexcept;

        R operator()(Args...) qualifiers;
        // `R`, `Args...`, and `qualifiers` are the return type, the parameter-type-
list,
        // and the sequence "cv-qualifier-seq-opt noexcept-specifier-opt" of the
function
        // type `Signature`, respectively.
    };

    template <typename Signature>
    constexpr void swap(function_ref<Signature>&, function_ref<Signature>&) noexcept;

    template <typename R, typename... Args>
    function_ref(R (*)(Args...)) -> function_ref<R(Args...)>;

    template <typename R, typename... Args>
    function_ref(R (*)(Args...) noexcept) -> function_ref<R(Args...) noexcept>;

    template <typename F>
    function_ref(F&&) -> function_ref<see below>;
}
```

The template argument `Signature` shall be a non-volatile-qualified function type.

<functional> header

Add the following to `[functional.syn]`:

```
namespace std
{
    // ...

    template <typename Signature> class function_ref;

    template <typename Signature>
    constexpr void swap(function_ref<Signature>& lhs, function_ref<Signature>& rhs)
    noexcept;

    // ...
}
```

Specification

```
template <typename Signature>
constexpr function_ref<Signature>::function_ref(const function_ref& rhs) noexcept;
```

- *Effects*: constructs a `function_ref` referring to the same callable `rhs` refers to.

```
template <typename Signature>
template <typename F>
constexpr function_ref<Signature>::function_ref(F&& f);
```

- *Requires*: none of the following must hold:
 - `f` is a null function pointer value.
 - `f` is a null member pointer value.

- F is an instance of the function class template, and $!f$.
- *Effects*: constructs a `function_ref` referring to f .
- *Remarks*: This constructor shall not participate in overload resolution unless `is_same_v<decay_t< $F$ >, function_ref>` is false and:
 - If Signature is marked `noexcept`: `is_nothrow_invocable_r_v< $R$ , cv-qualifiers  $F\&$ , Args...>`;
 - Otherwise: `is_invocable_r_v< $R$ , cv-qualifiers  $F\&$ , Args...>`.

Where R , $\text{Args} \dots$, and cv-qualifiers are the *return type*, the *parameter-type-list*, and the sequence “cv-qualifier-seq-opt” of the function type Signature, respectively.

```
template <typename Signature>
constexpr function_ref& function_ref<Signature>::operator=(const function_ref& rhs)
    noexcept;
```

- *Postconditions*: `*this` refers to the same callable `rhs` refers to.
- *Returns*: `*this`.

```
template <typename Signature>
template <typename F>
constexpr function_ref& function_ref<Signature>::operator=(F&&);
```

- *Requires*: none of the following must hold:
 - f is a null function pointer value.
 - f is a null member pointer value.
 - F is an instance of the function class template, and $!f$.
- *Remarks*: This function shall not participate in overload resolution unless `is_same_v<decay_t< $F$ >, function_ref>` is false and:
 - If Signature is marked `noexcept`: `is_nothrow_invocable_r_v< $R$ , cv-qualifiers  $F\&$ , Args...>`;
 - Otherwise: `is_invocable_r_v< $R$ , cv-qualifiers  $F\&$ , Args...>`.
- *Postconditions*: `*this` refers to f .

- *Returns:* `*this`.

```
template <typename Signature>
constexpr void function_ref<Signature>::swap(function_ref& rhs) noexcept;
```

- *Effects:* exchanges the values of `*this` and `rhs`.

```
template <typename Signature>
R function_ref<Signature>::operator()(Args... xs) qualifiers;
```

- *Effects:* equivalent to `return INVOKE<R>(f, std::forward<Args>(xs)...);`, where `f` is the callable object referred to by `*this`, qualified with the same *cv-qualifiers* as the function type `Signature`.

```
template <typename F>
function_ref(F&&) -> function_ref<see below>;
```

- *Remarks:* This deduction guide participates in overload resolution only if `&remove_reference_t<F>::operator()` is well-formed when treated as an unevaluated operand. In that case, if `decltype(&remove_reference_t<F>::operator())` is of the form `R(G::*)(A...) qualifiers` for a class type `G`, then the deduced type is `function_ref<R(A...) qualifiers>`, where `qualifiers` is the sequence “*cv-qualifier-seq-opt* *noexcept-specifier-opt*” of the function type `F`.

```
template <typename Signature>
constexpr void swap(function_ref<Signature>& lhs, function_ref<Signature>& rhs)
    noexcept;
```

- *Effects:* equivalent to `lhs.swap(rhs)`.

Feature test macro

I propose the feature-testing macro name `__cpp_lib_function_ref`.

Example implementation

An example implementation is available [here on GitHub](https://github.com/SuperV1234/Experiments/blob/master/function_ref.cpp) (https://github.com/SuperV1234/Experiments/blob/master/function_ref.cpp).

Existing practice

Many facilities similar to `function_ref` exist and are widely used in large codebases. Here are some examples:

- The `llvm::function_ref`² class template is used throughout LLVM. A quick GitHub search on the LLVM organization reports hundreds of usages both in `llvm` and `clang`³.
- Facebook’s Folly libraries⁴ provide a `folly::FunctionRef`⁵ class template. A GitHub search shows that it’s used in projects `proxygen` and `fbthrift`⁶.
- GNU’s popular debugger, `gdb`⁷, uses `gdb::function_view`⁸ throughout its code base. The documentation in the linked header file⁹ is particularly well-written and greatly motivates the need for this facility.

Additionally, combining results from GitHub searches (*excluding* “`llvm`” and “`folly`”) for “`function_ref`”¹⁰, “`function_view`”¹¹, “`FunctionRef`”¹², and “`FunctionView`”¹³ roughly shows more than 2800 occurrences.

Possible issues

Accepting temporaries in `function_ref`’s constructor is extremely useful in the most common use case: using it as a function parameter:

```
void foo(function_ref<void()>);
```

```
int main()
{
    foo([]{ });
}
```

(on [wandbox.org](https://wandbox.org/permlink/BPtPeQtErPGj4X7)) (<https://wandbox.org/permlink/BPtPeQtErPGj4X7>)

The usage shown above is completely safe: the temporary closure generated by the lambda expression is guaranteed to live for the entirety of the call to `foo`. Unfortunately, this also means that the following code snippet will result in *undefined behavior*:

```
int main()
{
    function_ref<void()> f{[]{ }};
    // ...
    f(); // undefined behavior
}
```

(on [wandbox.org](https://wandbox.org/permlink/cQPEX2sKjCQjgIki)) (<https://wandbox.org/permlink/cQPEX2sKjCQjgIki>)

The above closure is a temporary whose lifetime ends after the `function_ref` constructor call. The `function_ref` will store an address to a “dead” closure - invoking it will produce undefined behavior ¹⁴. As an example, `AddressSanitizer` detects an invalid memory access in this gist ¹⁵. Note that this problem is not unique to `function_ref`: the recently standardized `std::string_view` ¹⁶ has the same problem ¹⁷.

I strongly believe that accepting temporaries is a “necessary evil” for both `function_ref` and `std::string_view`, as it enables countless valid use cases. The problem of dangling references has been always present in the language - a more general solution like Herb Sutter and Neil Macintosh’s lifetime tracking ¹⁸ would prevent mistakes without limiting the usefulness of view/reference classes.

Open questions

Below are some unanswered questions for which I kindly ask guidance from members of the committee and readers of this paper.

- **(New in this revision)** `function_ref<Signature>`’s signature currently only accepts any combination of `const` and `noexcept`. Should this be extended to include *ref-qualifiers*? This would mean that `function_ref::operator()` would first cast the referenced callable to either an *lvalue reference* or *rvalue reference* (depending on `Signature`’s ref qualifiers) before invoking it. See P0045R1 ¹⁹ and N4159 ²⁰ for additional context.

- **(New in this revision)** Constructing a `std::function<Signature>` from a `function_ref<Signature>` is completely different from constructing a `std::string` from a `std::string_view`: the latter does actually create a copy while the former remains a reference. It may be reasonable to prevent implicit conversions from `function_ref` to `std::function` in order to avoid surprising dangerous behavior.
- `function_ref::operator()` is not currently marked as `constexpr` due to implementation issues. I could not figure a way to implement a `constexpr`-friendly `operator()`. Is there any possibility it could be marked as `constexpr` to increase the usefulness of `function_ref`?

Bikeshedding

The name `function_ref` is subject to bikeshedding. Here are some other potential names:

- `function_view`
- `callable_ref`
- `callable_view`
- `invocable_ref`
- `invocable_view`
- `fn_view`
- `fn_ref`

Acknowledgments

Thanks to **Agustín Bergé**, **Dietmar Kühl**, **Eric Niebler**, **Tim van Deurzen**, and **Alisdair Meredith** for providing very valuable feedback on earlier drafts of this proposal.

References

-
1. https://vittorioromeo.info/index/blog/passing_functions_to_functions.html#benchmark---generated-assembly.
(https://vittorioromeo.info/index/blog/passing_functions_to_functions.html#benchmark---generated-assembly)↩

2. http://llvm.org/doxygen/classllvm_1_1function_ref_3_01Ret_07Params_8_8_8_08_4.html
(http://llvm.org/doxygen/classllvm_1_1function_ref_3_01Ret_07Params_8_8_8_08_4.html)↵
3. https://github.com/search?q=org%3Allvm-mirror+function_ref&type=Code (https://github.com/search?q=org%3Allvm-mirror+function_ref&type=Code)↵
4. <https://github.com/facebook/folly> (<https://github.com/facebook/folly>)↵
5. <https://github.com/facebook/folly/blob/master/folly/Function.h#L743-L824>
(<https://github.com/facebook/folly/blob/master/folly/Function.h#L743-L824>)↵
6. <https://github.com/search?q=org%3Afacebook+FunctionRef&type=Code> (<https://github.com/search?q=org%3Afacebook+FunctionRef&type=Code>)↵
7. <https://www.gnu.org/software/gdb/> (<https://www.gnu.org/software/gdb/>)↵
8. <https://sourceware.org/git/gitweb.cgi?p=binutils-gdb.git;a=blob;f=gdb/common/function-view.h>
(<https://sourceware.org/git/gitweb.cgi?p=binutils-gdb.git;a=blob;f=gdb/common/function-view.h>)↵
9. <https://sourceware.org/git/gitweb.cgi?p=binutils-gdb.git;a=blob;f=gdb/common/function-view.h>
(<https://sourceware.org/git/gitweb.cgi?p=binutils-gdb.git;a=blob;f=gdb/common/function-view.h>)↵
10. https://github.com/search?utf8=%E2%9C%93&q=function_ref+AND+NOT+llvm+AND+NOT+folly+language%3AC%2B%2B&type=Code
(https://github.com/search?utf8=%E2%9C%93&q=function_ref+AND+NOT+llvm+AND+NOT+folly+language%3AC%2B%2B&type=Code)↵
11. https://github.com/search?utf8=%E2%9C%93&q=function_view+AND+NOT+llvm+AND+NOT+folly+language%3AC%2B%2B&type=Code
(https://github.com/search?utf8=%E2%9C%93&q=function_view+AND+NOT+llvm+AND+NOT+folly+language%3AC%2B%2B&type=Code)↵
12. <https://github.com/search?utf8=%E2%9C%93&q=functionref+AND+NOT+llvm+AND+NOT+folly+language%3AC%2B%2B&type=Code>
(<https://github.com/search?utf8=%E2%9C%93&q=functionref+AND+NOT+llvm+AND+NOT+folly+language%3AC%2B%2B&type=Code>)↵
13. <https://github.com/search?utf8=%E2%9C%93&q=functionview+AND+NOT+llvm+AND+NOT+folly+language%3AC%2B%2B&type=Code>
(<https://github.com/search?utf8=%E2%9C%93&q=functionview+AND+NOT+llvm+AND+NOT+folly+language%3AC%2B%2B&type=Code>)↵
14. <http://foonathan.net/blog/2017/01/20/function-ref-implementation.html>
(<http://foonathan.net/blog/2017/01/20/function-ref-implementation.html>)↵

15. <https://gist.github.com/SuperV1234/a41eb1c825bfbb43f595b13bd4ea99c3>
(<https://gist.github.com/SuperV1234/a41eb1c825bfbb43f595b13bd4ea99c3>)↵
16. <http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2013/n3762.html> (<http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2013/n3762.html>)↵
17. http://foonathan.net/blog/2017/03/22/string_view-temporary.html (http://foonathan.net/blog/2017/03/22/string_view-temporary.html)↵
18. <https://github.com/isocpp/CppCoreGuidelines/blob/master/docs/Lifetimes%20I%20and%20II%20-%20v0.9.1.pdf>
(<https://github.com/isocpp/CppCoreGuidelines/blob/master/docs/Lifetimes%20I%20and%20II%20-%20v0.9.1.pdf>)↵
19. <http://wg21.link/p0045r1> (<http://wg21.link/p0045r1>)↵
20. <http://wg21.link/N4159> (<http://wg21.link/N4159>)↵